CS211 PASS MOCK MST S1 2025

1. List the major highlight in terms of technology used to build computers for each of the following computing generations.(4 marks)
2. Suppose you design a computer with a ten stage pipeline to execute one instruction with each stage taking 5 nsec. How long will it take to execute a program that has 30 sequential instructions? (2 marks)
3. The CPU consists of a control unit, arithmetic unit, and registers. Explain the purpose of the control unit and arithmetic unit. (3 marks)
4. Explain the reason for designing a new set of instructions in a multi level machine. (2 marks)
5. List two differences between RISC and CISC processor architecture. (2marks)
6. What would be the codeword if you would encode the data word 100111000010 to allow one but error detection and correction using hamming code with even parity? (3 marks)
7. Convert -22.5 to IEEE sincle-precision format. Represent the final answer in hexadecimal. (3 marks)
8. Find the sum of 14 and -15 in 8 bits binary using ones complement and twos complement. (2 marks)
9. A magnetic disk has 400 cylinders, 15 tracks per cylinder, 128 sectors per track and 2 KB per sector. The rotation time is 20 msec, maximum seek time is 750 msec, and switching between adjacent cylinders takes 35 msec. What is the average access time to read 10 MB of data stored consecutively on your disk? (3 marks)
10. Represent how -15 will be stored in binary in the RAM of the CPUSIM machine. (2 marks)
11. Suppose you want to store the decimal values 15 in machine that follows big endian byte ordering.
    a. Show the contents of the memory in hexadecimal. Assume the number will be stored at address 12. (2 marks)
    b. Now suppose you want to transfer the contents of address 12 above to a machine that follows little endian byte ordering. Show the contents of the memory in hexadecimal. (2 marks)

9. Suppose you want to store the decimal value 15 in machine that follows big endian byte ordering.
   a. Show the contents of the memory in hexadecimal. Assume the number will be stored at address 12. (2 marks)

12

00	00	00	0F

   b. Now suppose you want to transfer the contents of address 12 above to a machine that follows little endian byte ordering. Show the contents of the memory in hexadecimal. (2 marks)

0F	00	00	00

7. Represent how -15 will be stored in binary in the RAM of a CPUSIM machine. (2 marks)

1111 1111 1111 0001

5. List the major highlight in terms of technology used to build computers for each of the following computing generations. (4 marks)

Zeroth Generation _____Mechanical Computers_____

First Generation _____Vacuum Tubes _____

Second Generation _____Transistors_____

Third Generation _____Integrated Circuits_____

1 mark for each

6. Explain the purpose and concept of pipelining. (2 marks)

Pipelining is instruction level parallelism, which improves performance of the machine. An instruction divided into many (often a dozen or more) parts, each one handled by a dedicated piece of hardware, all of which can run in parallel.

2 marks for correct solution. 1 mark for partially correct solution.

7. Suppose you design a computer with a ten stge pipeline to execute one instruction, with each stage taking 5 nsec. How long will it take to execute a program that has 30 sequential instructions? (2 marks)

Instruction 1 : 10 cycles

Remaining 29: 29 cycles

Total: 39 cycles X 5 = 195 nsec

8. The CPU consists of the control unit, arithmetic unit, and registers. Explain the purpose of the control unit and arithmetic unit. (3 marks)

The control unit is responsible for fetching instructions from main memory and determining their type. (1½ m)

The arithmetic logic unit performs operations such as addition and Boolean AND needed to carry out the instructions. (1½ m)

1. Explain the reason for designing new set of instructions in a multi-level machine. (2 marks)

To bridge the gap between what is convenient for people and what is convenient for computers.

2 marks. No partial marks

6. Represent the decimal value 12 in hexadecimal in a 32 bit big endian and little endian machines at address 4. (3 marks)

00	00	00	0C

The number representation will be the same except that the address will be 4 on high end for big endian and low end for little endian machine.

9) What would be the codeword if you would encode the data word 100111000010 to allow one bit error detection and correction using hamming code with even parity? (3 marks)

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
$\boxed{0}$	$\boxed{0}$	1	$\boxed{0}$	0	0	1	$\boxed{1}$	1	1	0	0	0	0	1	$\boxed{0}$	0

Bit 1: 1, 3, 5, 7, 9, 11, 13, 15, 17 — 4

2: 2, 3, 6, 7, 10, 11, 14, 15 — 4

4: 4, 5, 6, 7, 12, 13, 14, 15 — 2

8: 8, 9, 10, 11, 12, 13, 14, 15 — 3

16: 16, 17 — 0

$2\frac{1}{2}$

∴ codeword = 0010 0011 1100 00100 $\}\frac{1}{2}$

3. List two differences between RISC and CISC processor architectures. (2 marks)

CISC is more complex / RISC is simpler
CISC longer instruction set/ RISC shorter instruction set
RISC requires more RAM
CISC many address modes / RISC has less or 1 mode
CISC may have more registers/ RISC has plenty registers
RISC takes one machine cycle / CISC takes many cycles to complete one instruction
RISC fixed number of bytes / CISC variable number

1. What is the hamming distance between 1010 1010 and 0101 0101? (1 mark)

8

8) Convert -22.5 to IEEE single-precision format. Represent the final answer in hexadecimal. (3 marks)

22 ⇒ 10110
0.5 ⇒ 0.1] ½ mark

0.5
× 2
1 | .0

22.5 ⇒ 10110.1
Normalise: 1.01101 × 2^4 } ½ mark

Stored exp = actual + excess
= 4 + 127
= 131
⇒ 10 00 0011 } 1 mark

1 | 10 00 0011 | 011 0100 0 . . . . . 0 ½ mark
sign | exp

C 1 B 4 0 0 0 0 ½ mark.

9) What would be the codeword if you would encode the data word 100111000010 to allow one

11) Find the sum of 14 and -15 in 8 bits binary using One's complement and Two's complement (2 marks)

1 mark

One's	Two's
14 = 0000 1110	14 = 0000 1110
-15 = + 1111 0000	-15 = 1111 0001
1111 1110	1111 1111

1 mark

10. A magnetic disk has 400 cylinders, 15 tracks per cylinder, 128 sectors per track and 2 KB per sector. The rotation time is 20 msec, maximum seek time is 750 msec, and switching between adjacent cylinders takes 35 msec. What is the average access time to read 10 MB of data stored consecutively on your disk? (3 marks)

Transfer time:

$$10\ \cancel{MB} \times \frac{1024\ \cancel{KB}}{1\ \cancel{MB}} \times \frac{1\ \cancel{sec}}{2\ \cancel{KB}} \times \frac{1\ track}{128\ \cancel{sec}} = 40\ tra$$

Transfer time = 40 × 20
= 800 msec (1)

Avg s/w: 80×6×1/15 + 120×9×1/15 = 104 msec (1)

Track	Prob.	#switch s/w	
1	1/15	2	80
2	⋮	2	
3	⋮	2	
4	⋮	2	
5	⋮	2	
6	⋮	2	
7	⋮	3	120
8	⋮	3	
9	⋮	3	
10	⋮	3	
11	⋮	3	
12	⋮	3	
13	⋮	3	
14	⋮	3	
15	1/15	3	

∴ Avg. access time:
avg seek + avg lat + transfer + avg s/w
(1) = (1/3×750) + (1/2×20) + 800 + 104
= 1164 msec